



LESSON PLAN

FOR

Data Mining and Machine Learning Fundamentals for Beginners

Course Code: TGS-2020503264

Conducted by Tertiary Infotech Academy Pte Ltd

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Version 17.0 · 10 September 2026

Document Version Control Record

Version	Effective date	Summary of changes	Author
17.0	10 September 2026	Replanned two-day delivery around the central Python/Streamlit app, 15 aligned labs, and the current WA and PP assessment pattern.	Dr Alfred Ang

Course Information

Course	Data Mining and Machine Learning Fundamentals for Beginners
Course code	TGS-2020503264
Duration	2 days · 8 training hours per day
Assessment	WA 1 hour + PP 1 hour, open book
Training mode	Instructor-led concepts, demonstrations, guided practice, and evidence review

Learning Outcomes

- LO1: Apply data mining and machine learning principles to assess business insights.
- LO2: Integrate and aggregate information from multiple datasets to build data models.
- LO3: Apply predictive data modelling techniques to identify underlying trends in data.
- LO4: Apply machine learning classification techniques to gain new insights from data.
- LO5: Apply clustering techniques to discover patterns, visualise evidence, and make decisions.
- LO6: Develop prototype algorithms with dimension-reduction and feature-ranking techniques.
- LO7: Construct association rules to identify patterns across transactional datasets.

Detailed Schedule

Day 1

Time	Duration	Type	Learning activity and evidence
09:00–09:30	30 min	Admin	Welcome, learning outcomes, digital attendance and evidence expectations
09:30–10:45	75 min	Topic 1	Business framing, data-mining evidence loop, supervised and unsupervised boundaries
10:45–11:00	15 min	Break	Tea break
11:00–13:00	120 min	Topic 2 + Labs	Data contracts, data quality, joins, imputation; Labs 1–3
13:00–14:00	60 min	Lunch	Lunch break
14:00–15:30	90 min	Topic 2/3 + Labs	Scaling, aggregation, regression mechanism and metrics; Labs 4–5
15:30–15:45	15 min	Break	Tea break
15:45–17:45	120 min	Topic 3/4 + Labs	Regularisation, classification models, cross-validation; Labs 6–8
17:45–18:00	15 min	Recap	PM digital attendance, evidence review and questions

Total training/assessment time excluding lunch: 480 minutes (8 hours).

Day 2

Time	Duration	Type	Learning activity and evidence
09:00–09:15	15 min	Recap	AM digital attendance and Day 1 retrieval practice
09:15–10:45	90 min	Topic 4 + Lab	Confusion matrix, ROC/AUC and threshold policy; Lab 9
10:45–11:00	15 min	Break	Tea break
11:00–13:00	120 min	Topic 5 + Labs	K-means, silhouette and hierarchy; Labs 10–11
13:00–14:00	60 min	Lunch	Lunch break
14:00–15:30	90 min	Topics 6/7 + Labs	PCA, feature ranking and association rules; Labs 12–14
15:30–15:45	15 min	Break	Tea break
15:45–16:00	15 min	Lab + Briefing	Lab 15 anomaly evidence and assessment briefing
16:00–17:00	60 min	Assessment	Written Assessment (WA), open book
17:00–18:00	60 min	Assessment	Practical Performance (PP), open book; PM/assessment attendance

Total training/assessment time excluding lunch: 480 minutes (8 hours).